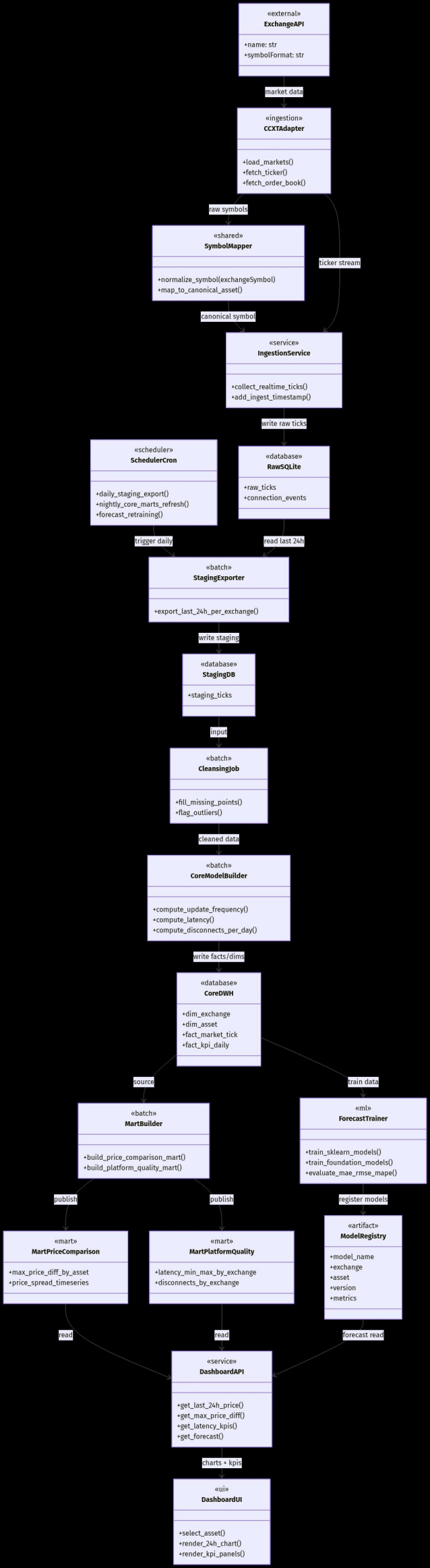
«external»
ExchangeAPI
+name: str
+symbolFormat: str
market data
«ingestion»
CCXTAdapter
+load_markets()
+fetch_ticker()
+fetch_order_book()
raw symbols
ticker stream
«shared»
SymbolMapper
+normalize_symbol(exchangeSymbol)
+map_to_canonical_asset()
canonical symbol
«service»
IngestionService
+collect_realtime_ticks()
+add_ingest_timestamp()
write raw ticks
«scheduler»
SchedulerCron
+daily_staging_export()
+nightly_core_marts_refresh()
+forecast_retraining()
«database»
RawSQLite
+raw_ticks
+connection_events
trigger daily
read last 24h
«batch»
StagingExporter
+export_last_24h_per_exchange()
write staging
«database»
StagingDB
+staging_ticks
input
«batch»
CleansingJob
+fill_missing_points()
+flag_outliers()
cleaned data
«batch»
CoreModelBuilder
+compute_update_frequency()
+compute_latency()
+compute_disconnects_per_day()
write facts/dims
«database»
CoreDWH
+dim_exchange
+dim_asset
+fact_market_tick
+fact_kpi_daily
source
train data
«batch»
MartBuilder
+build_price_comparison_mart()
+build_platform_quality_mart()
«ml»
ForecastTrainer
+train_sklearn_models()
+train_foundation_models()
+evaluate_mae_rmse_mape()
publish
publish
register models
«mart»
MartPriceComparison
+max_price_diff_by_asset
+price_spread_timeseries
«mart»
MartPlatformQuality
+latency_min_max_by_exchange
+disconnects_by_exchange
«artifact»
ModelRegistry
+model_name
+exchange
+asset
+version
+metrics
read
read
forecast read
«service»
DashboardAPI
+get_last_24h_price()
+get_max_price_diff()
+get_latency_kpis()
+get_forecast()
charts + kpis
«ui»
DashboardUI
+select_asset()
+render_24h_chart()
+render_kpi_panels()